

CrossSampleCNP

Conditional Neural Process · Protein Viscosity Predictor

ENCODING

CONTEXT PATH

QUERY PATH

AGGREGATION

DECODING

LEGEND

- Input / passthrough
- Encoder MLP
- Attention Pooler
- Latent representation r
- Decoder MLP
- Output

$D = \text{static_dim} = 85$ $H = \text{hidden_dim} = 128$ $L = \text{latent_dim} = 18$ $[B \times N_q \times 1]$ one value per shear rate per query

Context Input
concentration · log η · static features $[B \times N_{\text{ctx}} \times (2 + D)]$

Encoder MLP
Linear $(2+D, H) \rightarrow \text{ReLU}$
Linear $(H, H) \rightarrow \text{ReLU}$
Linear $(2+D \rightarrow H) \cdot \text{ReLU} \cdot \text{Dropout}$
Linear $(H, \text{latent_dim})$

Attention Pooler
Learned Query $q \in \mathbb{R}^{\text{latent}}$
MultiheadAttention (Q, K, V)
LayerNorm
Learnable query $n_{\text{heads}} = 4$

Latent Context r
 $[B \times \text{latent_dim}]$ · protein-level summary

broadcast
across queries

Query Input
shear rate · static features $[B \times N_q \times (1 + D)]$

Query Features (unchanged)
shear rate · static $[B \times N_q \times (1+D)]$

Concatenate
 $r_{\text{expanded}} [B \times N_q \times L]$ $query [B \times N_q \times (1+D)]$
shear || static || r_{expanded} $[B \times N_q \times (1+D+\text{latent_dim})]$

Decoder MLP
Linear $(1+D+L, H) \rightarrow \text{ReLU}$
Linear $(H, H) \rightarrow \text{ReLU}$
Linear $(1+D+L \rightarrow H) \cdot \text{ReLU} \cdot \text{Dropout}$
Linear $(H, 1)$

log $\hat{\eta}$ (predicted log-viscosity)

N_{ctx} = number of context observations · N_q = number of query shear rates · B = batch size